Concepts of Biostatistics and Data Science

A two-day crash course with practical analysis in R

Day 1: Understanding data and making comparisons



Plan for today

Time	Session	Focus
09:00 to 09:10	Welcome	Aims, data, how the course works
09:10 to 09:30	Theory 1	Questions, observations, variables, data quality
09:30 to 09:55	Theory 2	Describing data: summaries and plots
09:55 to 10:20	Theory 3	Samples, variability, effect sizes, CIs, tests
10:20 to 10:42	Theory 4	Comparing groups: continuous outcomes
10:42 to 11:00	Theory 5	Categorical associations; first look at confounding
11:00 to 11:30	Break	(not counted in teaching time)
11:30 to 13:30	Practical	Import, inspect, describe, compare in R

By the end of today you will be able to

- Describe a dataset in terms of observations, variables and variable types
- Check a dataset for missing values, impossible values and coding problems
- Choose and calculate suitable summaries: counts, proportions, mean (SD), median (IQR)
- Choose a suitable plot: bar chart, histogram or boxplot
- Explain what a confidence interval and a p-value do and do not tell you
- Compare two or more groups with a t-test, ANOVA or a rank-based test, and say why
- Analyse a contingency table and explain proportions, risks, odds, RR and OR
- Recognise paired data and avoid analysing it as independent groups

Our data: the Healthy Hearts District Study (simulated)

What it contains

- 480 adults aged 30 to 74 from four districts
- Baseline: age, sex, education, smoking, activity, diet score, BMI, blood pressure, CRP
- Randomised: standard advice or a lifestyle programme
- 12-month follow-up: blood pressure, satisfaction, adverse events, clinic visits

Why we use it

- Simulated with known rules, so results are for teaching only
- Realistic problems built in: missing values, a -99 code, skewed variables, people who left early
- Data dictionary: data/data_dictionary.csv
- One dataset all course: you learn the methods, not new data each time



Questions, observations and variables

Part 1. Analysis starts with a question, not with a test

Start with the question

Question	Purpose	What we will need
What proportion of participants smoke?	Describe	Count and proportion
Is mean blood pressure higher in men than in women?	Compare groups	Difference in means with CI
Is smoking associated with high blood pressure?	Association	Table, RR or OR with CI
Did blood pressure change after the programme?	Compare paired measurements	Mean change with CI

> The method follows from the question, the type of outcome and the study design.

Observations and variables

- An observation is one unit we measured: here, one participant (one row)
- A variable is one characteristic recorded for every observation (one column)
- Before analysing, ask: what does one row represent?
  - If a person is measured twice, the two values are linked (paired)
  - If people come from the same clinic or family, they may be more alike (clustered)
- Standard methods assume each observation is independent of the others

Types of variables

Type	Meaning	Example in our data
Continuous	Any value on a scale	sbp_0, bmi, crp, age
Count (discrete)	Whole numbers 0, 1, 2 ...	clinic_visits
Binary	Two categories	smoker, high_bp_0, adverse_event
Nominal	Categories without order	district
Ordinal	Ordered categories; gaps may not be equal	education, satisfaction (1 to 5)
Time-to-event	Time until something happens, often censored	not in our data (further learning)

Check data quality before any analysis

- Range: are values possible? (activity of -99 minutes is a code, not a value)
- Missing values: how many, in which variables, and why?
  - 41 people have no 12-month blood pressure because they left the study early
- Categories: consistent spelling and labels ("North", "north", "N")
- Duplicates and identifiers: one row per person?
- Types: is each variable stored as a number or a category, as intended?

> Most R functions silently drop rows with missing values. Always report how many observations each result uses.



Describing data

Part 2. Summaries and plots that suit the variable

Categorical variables: counts and proportions

- A proportion is a count divided by a denominator: always say what the denominator is
  - Smokers: 79 / 480 = 16.5% of all participants
  - Tertiary education: 134 / 474 = 28.3% of those with education recorded (6 missing)
- Percentages without counts can mislead when groups are small
- Plot with a bar chart: one bar per category, bars separated by gaps

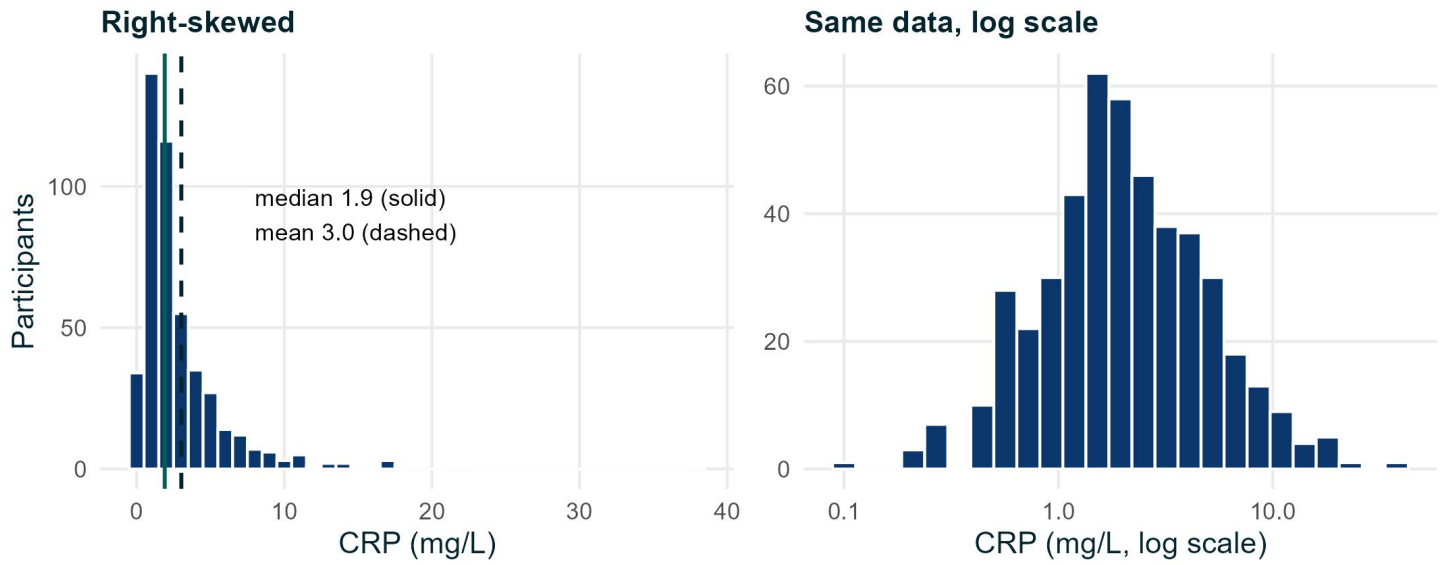
Centre and spread for continuous variables

Summary	What it describes	SBP at baseline (n = 480)
Mean	Average value; sensitive to extreme values	129.3 mmHg
Standard deviation (SD)	Typical distance of values from the mean	15.6 mmHg
Median	Middle value when sorted	129 mmHg
Interquartile range (IQR)	Range of the middle 50% (25th to 75th percentile)	118 to 140 mmHg

- Symmetric distribution: mean (SD) and median (IQR) tell a similar story
- Skewed distribution: report median (IQR), or analyse on a log scale

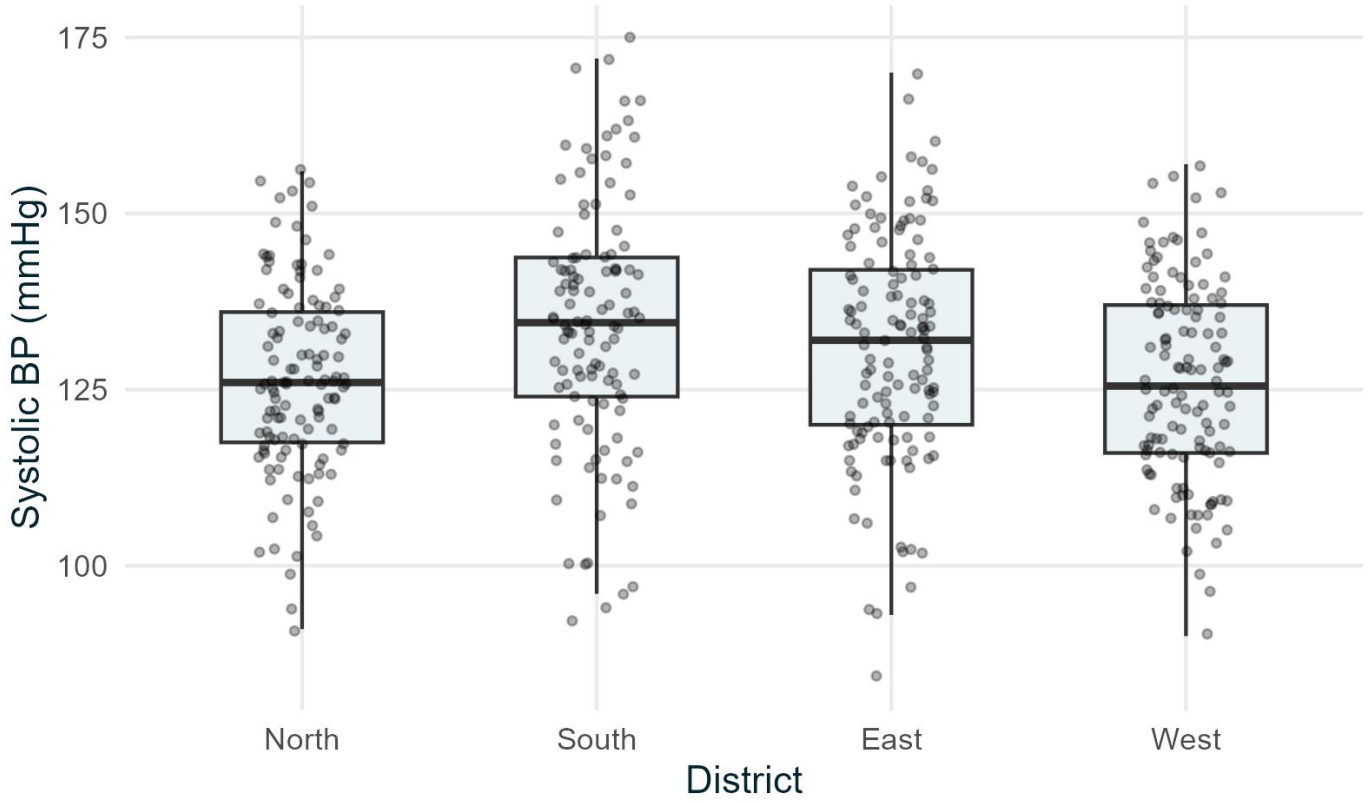
Skewed data: C-reactive protein

- Most values are low; a few are very high
- Mean 3.0 mg/L, median 1.9 mg/L
- SD (3.4) is larger than the mean: a sign of right skew
- Report median (IQR): 1.9 (1.2 to 3.7) mg/L
- On a log scale the distribution is roughly symmetric



Choosing a plot

- One categorical variable: bar chart
- One continuous variable: histogram (shape) or boxplot (summary)
- Continuous by group: boxplots with the points shown
- Two continuous variables: scatterplot (Day 2)
  - Box: IQR; line: median; whiskers: up to 1.5 IQR





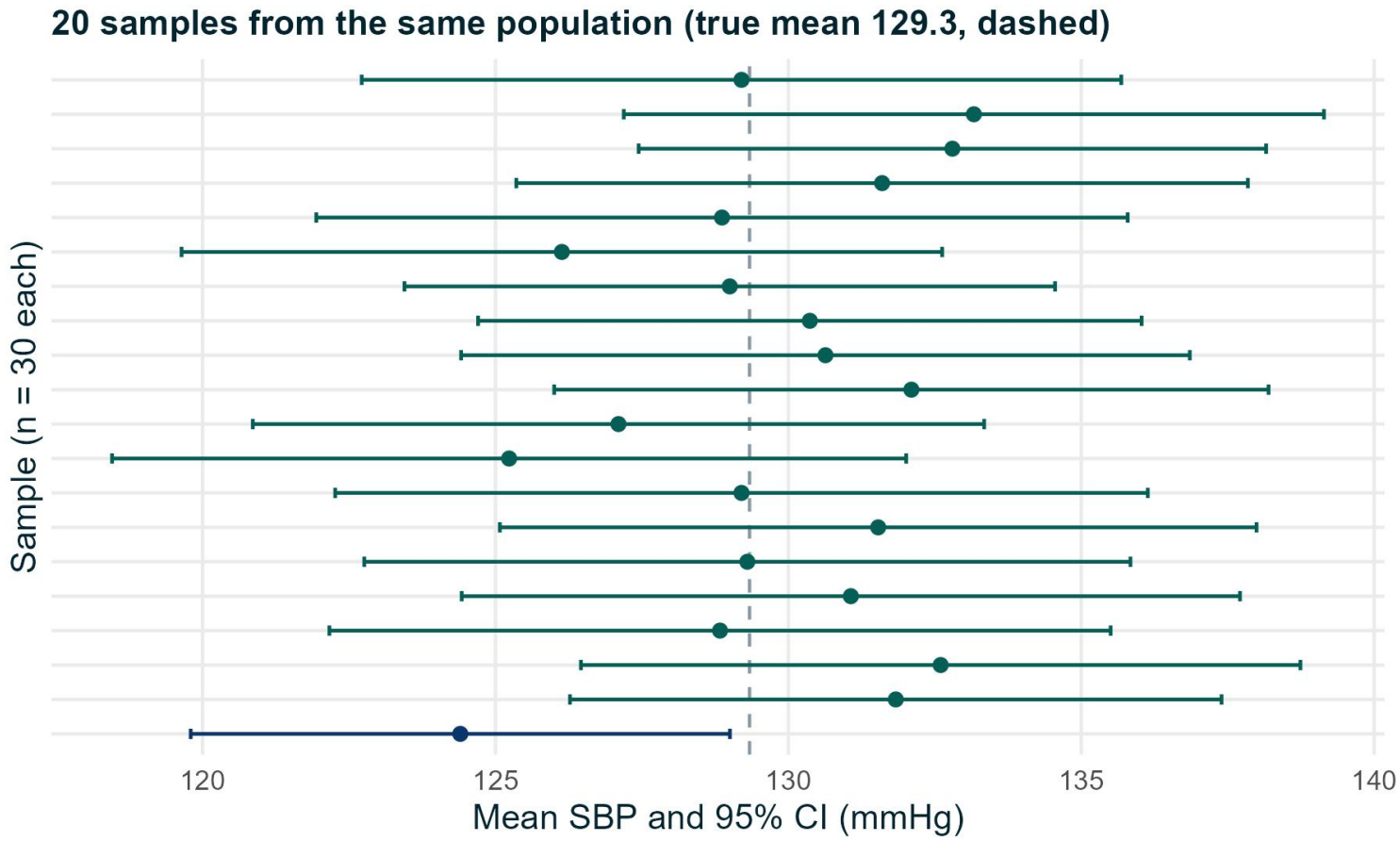
From a sample to a population

Part 3. Variability, effect sizes, confidence intervals and tests

Samples and populations

- Population: everyone we want to draw conclusions about
- Sample: the people we actually measured
- A statistic (e.g. a sample mean) estimates a population value
- A different sample would give a different estimate: this is sampling variability
- Statistics quantifies that uncertainty; it cannot remove bias from a poor sample

The same study repeated 20 times



Each line is one sample of 30 people with its 95% CI. 19 of the 20 intervals include the true mean.

Standard error and confidence interval

- SD describes how individuals vary; the standard error (SE) describes how an estimate varies
- For a mean: SE = SD / √n. Larger samples give smaller SEs
- 95% confidence interval ≈ estimate ± 2 × SE
- Interpretation: the range of population values reasonably compatible with the data
  - Strictly: the method captures the true value in 95% of repeated samples
- Example: mean SBP 129.3 mmHg, SE = 15.6 / √480 = 0.71, 95% CI about 127.9 to 130.7

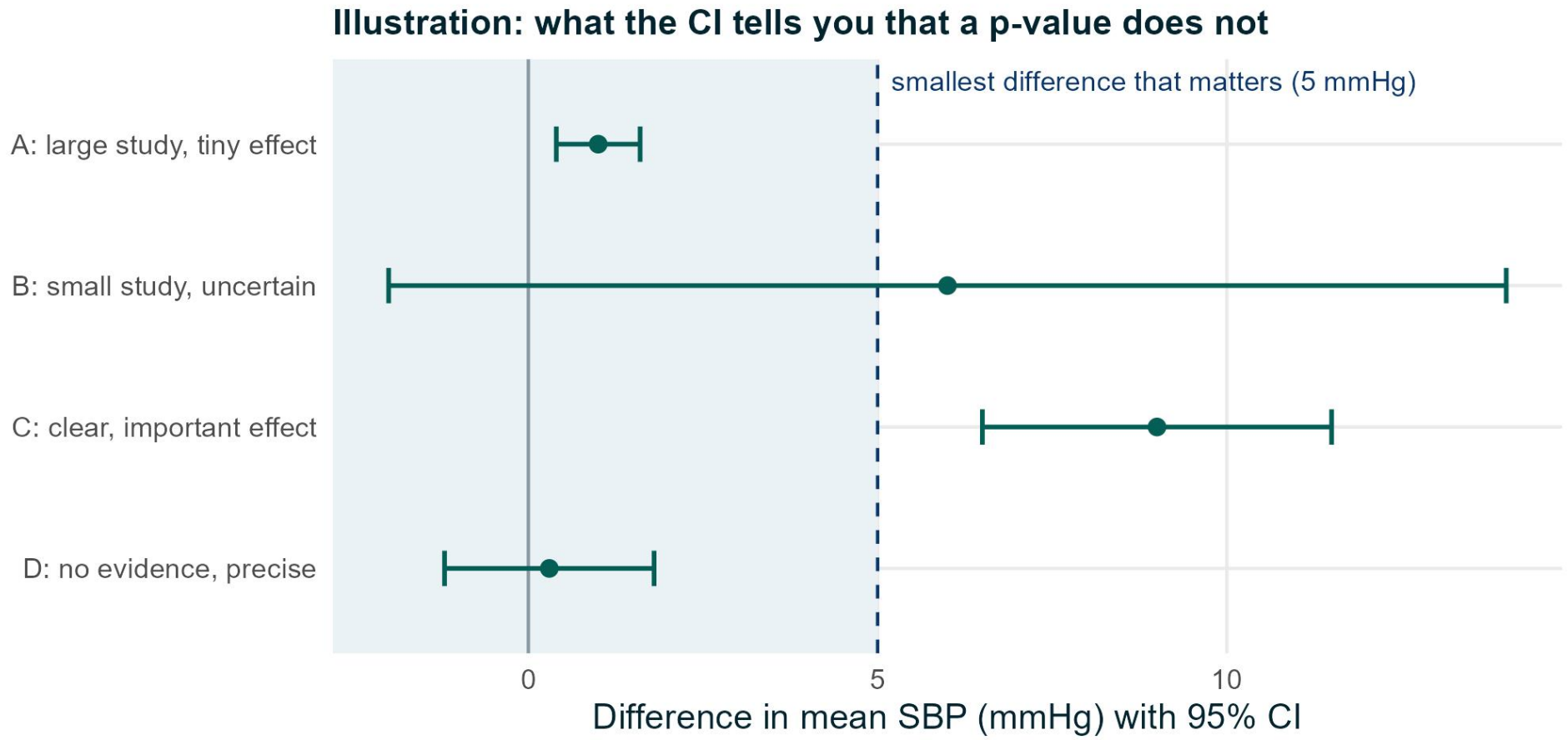
Effect sizes: how big is the difference?

- An effect size measures the size of a difference or association in meaningful units
  - Difference in means: men's SBP is 6.1 mmHg higher than women's
  - Ratio of geometric means: smokers' CRP is 1.6 times that of non-smokers
  - Difference or ratio of proportions, odds ratio
- Always report an effect size with its 95% CI
- A p-value alone says nothing about the size of an effect

Hypothesis tests and p-values

- Null hypothesis (H0): no difference or no association in the population
- p-value: if H0 were true, the probability of a result at least as extreme as ours
- A small p-value means the data are hard to reconcile with H0
- The p-value is NOT the probability that H0 is true
- p > 0.05 is NOT evidence of no difference: look at the CI
- 0.05 is a convention, not a law. Report the actual p-value (e.g. p = 0.03)

Statistical significance is not practical importance



Hypothetical studies. Judge results by the estimate and both ends of the CI.

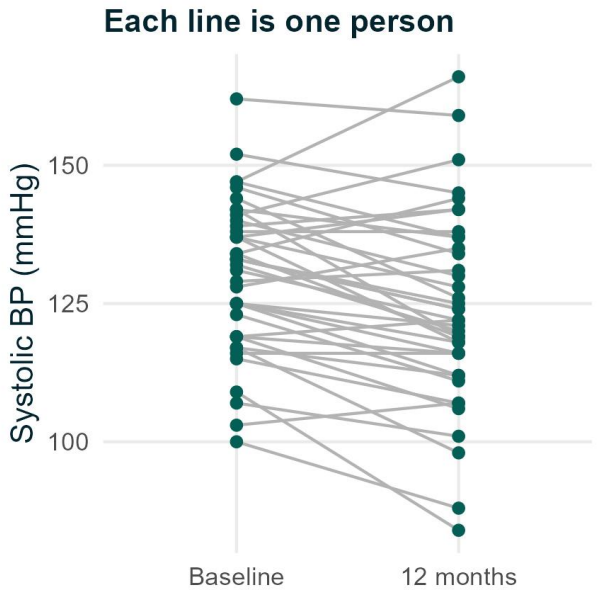


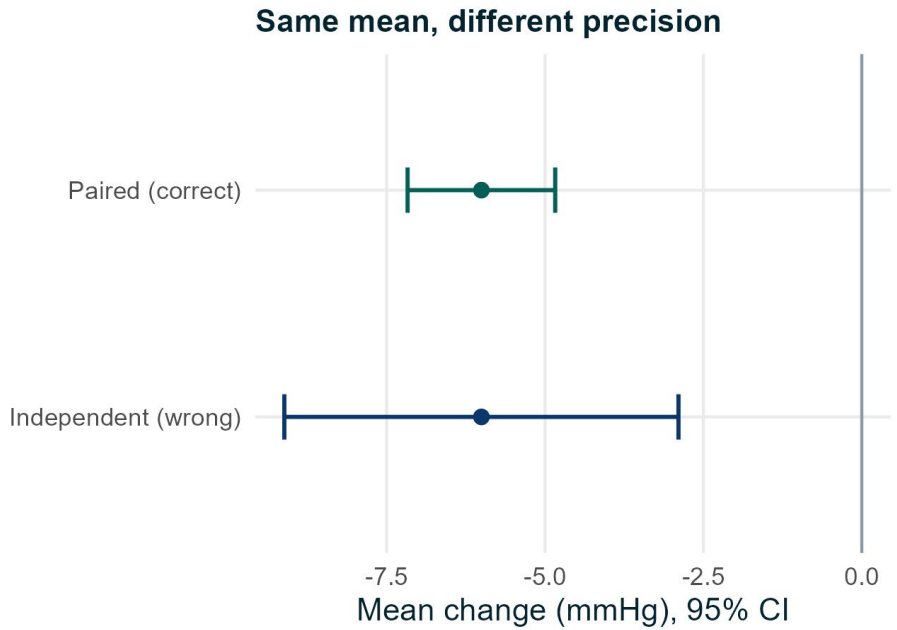
Comparing groups: continuous outcomes

Part 4. Two groups, more than two groups, and paired data

First question: independent or paired?

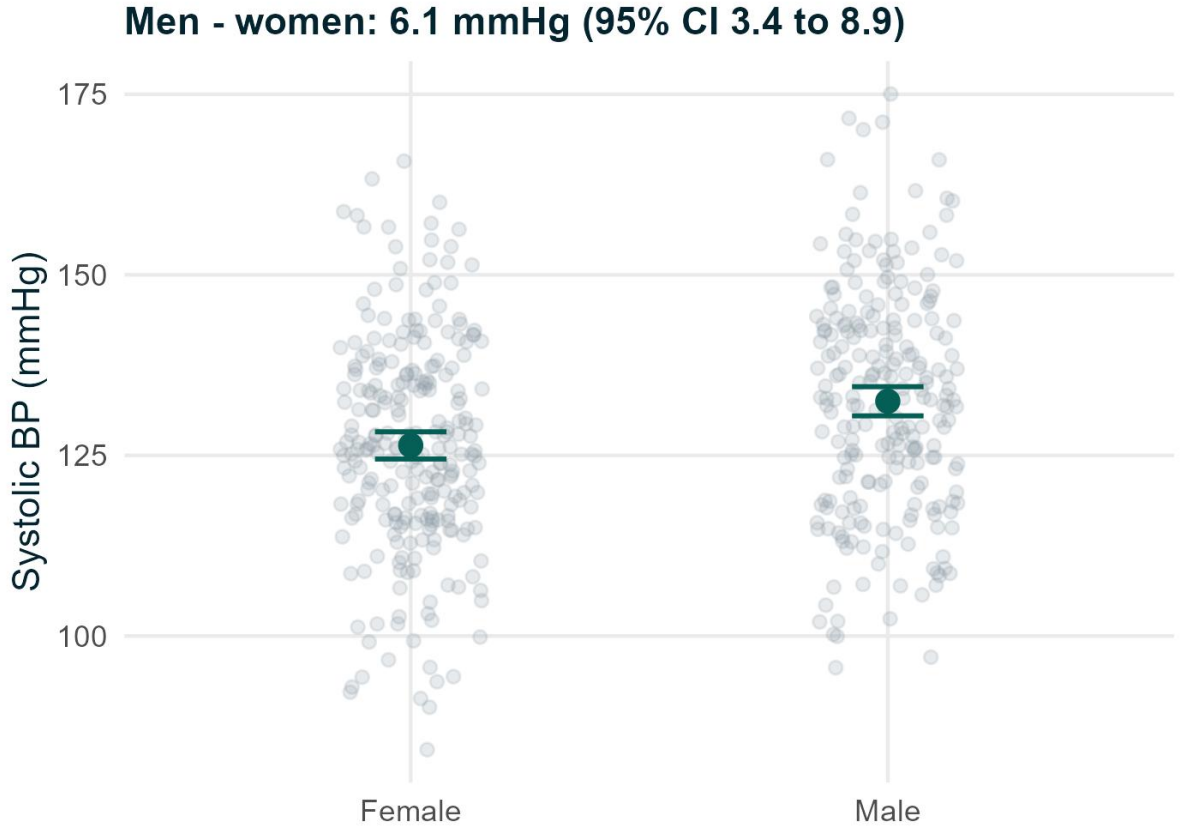
- Independent: different people in each group (men vs women)
- Paired: the same people measured twice, or matched pairs
- Paired designs remove between-person differences, so they are more precise
- Analysing paired data as independent gives the wrong CI and p-value





Two independent groups: Welch's t-test

- Question: does mean SBP differ between men and women?
- Difference (men - women): 6.1 mmHg, 95% CI 3.4 to 8.9, p < 0.001
- Welch's version does not assume equal variances: use it by default
- Assumptions: independent observations; sample means approximately normal
  - Holds for large groups even if data are moderately skewed



A skewed outcome: CRP in smokers and non-smokers

Option 1: log scale

- Analyse log(CRP) with a t-test, then back-transform
- Ratio of geometric means: 1.63 (95% CI 1.31 to 2.03)
- "Typical CRP is about 60% higher in smokers"

Option 2: rank-based

- Wilcoxon rank-sum (Mann-Whitney) test, $p < 0.001$
- Asks whether values in one group tend to be larger
- Hodges-Lehmann shift 0.9 mg/L (0.5 to 1.4): not a difference in medians

> Rank tests still assume independent observations. They are not assumption-free and not, in general, tests of medians.

Choosing between t-tests and rank-based tests

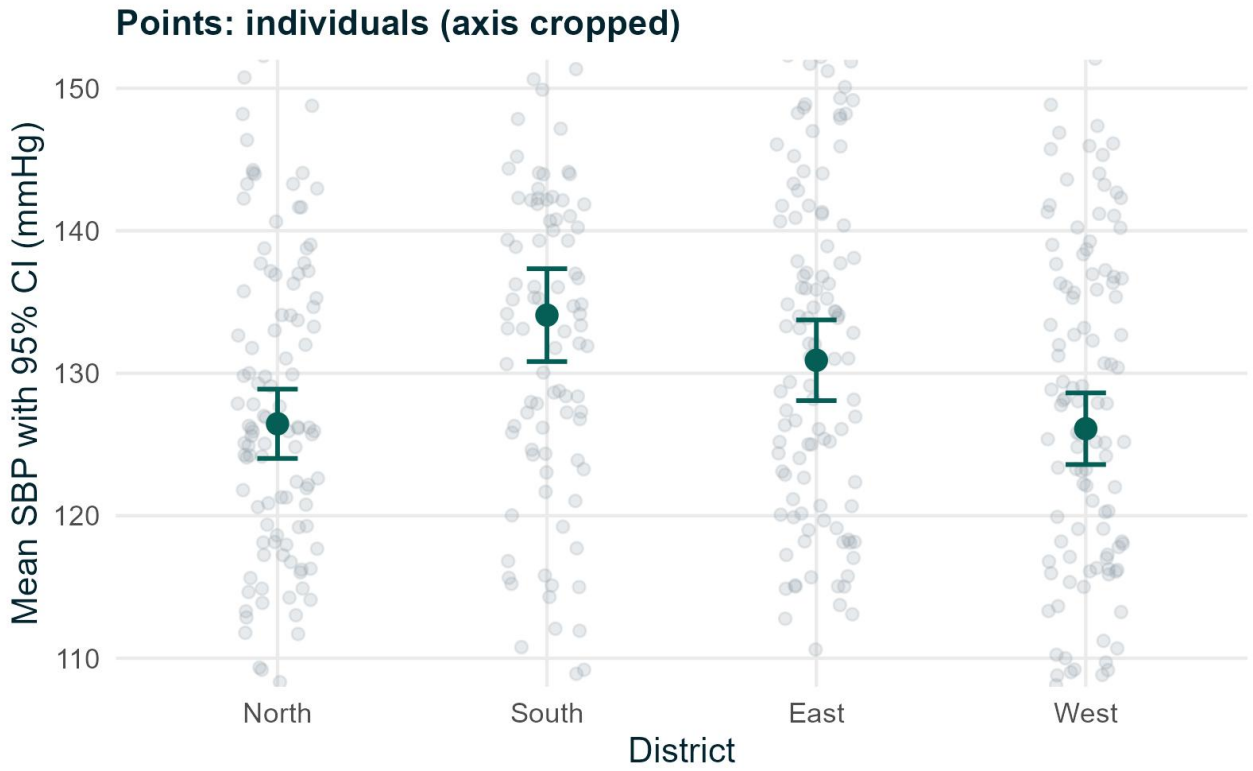
- Ask first: what summary answers the question? Mean, typical (geometric mean), or tendency to be larger?
- Look at plots of the data, not only a normality-test p-value
  - Normality tests reject trivial departures in large samples and miss big ones in small samples
- A small sample does not automatically require a nonparametric test
  - With few observations every method has little information; rank tests can have very low power
- For t-tests the normality assumption concerns the sampling distribution of the mean (or the residuals)

Paired data: the paired t-test

- Calculate each person's change, then analyse the changes
- Lifestyle group (n = 224): mean change -6.0 mmHg, 95% CI -7.2 to -4.8
- Standard advice group (n = 215): mean change -2.0 mmHg, 95% CI -3.2 to -0.8
- Rank-based alternative: Wilcoxon signed-rank test
- To compare the programme with standard advice, compare the changes between the two groups:
  - difference 4.0 mmHg, 95% CI 2.3 to 5.6 (Welch t-test on the changes)

More than two groups: one-way ANOVA

- Question: does mean SBP differ between the four districts?
- Welch's ANOVA: $F = 6.8$, $p < 0.001$
- The overall test says the means are not all equal
- It does not say which districts differ
- Assumptions: independence, similar spread (or use Welch), approximately normal residuals



Which groups differ? Pairwise comparisons

Comparison (Tukey)	Difference (mmHg)	95% CI (adjusted)
South - North	7.6	2.5 to 12.8
East - North	4.5	-0.6 to 9.5
West - North	-0.3	-5.4 to 4.7
East - South	-3.2	-8.3 to 1.9
West - South	-8.0	-13.1 to -2.8
West - East	-4.8	-9.8 to 0.2

- Six comparisons: adjust for multiple testing (Tukey, or Holm with pairwise.t.test)
- Better still: decide the comparisons that matter before seeing the data

Kruskal-Wallis: a rank-based comparison of several groups

- Extends the Wilcoxon rank-sum test to three or more independent groups
- Asks whether values tend to be larger in some groups than others
- CRP by district: medians 1.8 to 2.5 mg/L, Kruskal-Wallis p = 0.07
- A p-value above 0.05 is not proof of no difference: the data are inconclusive
- Like Wilcoxon, it assumes independence and is not generally a test of medians
- Follow-up: pairwise Wilcoxon tests with adjusted p-values, if justified



Associations between categorical variables

Part 5. Contingency tables, risks, odds, chi-square and Fisher's exact test

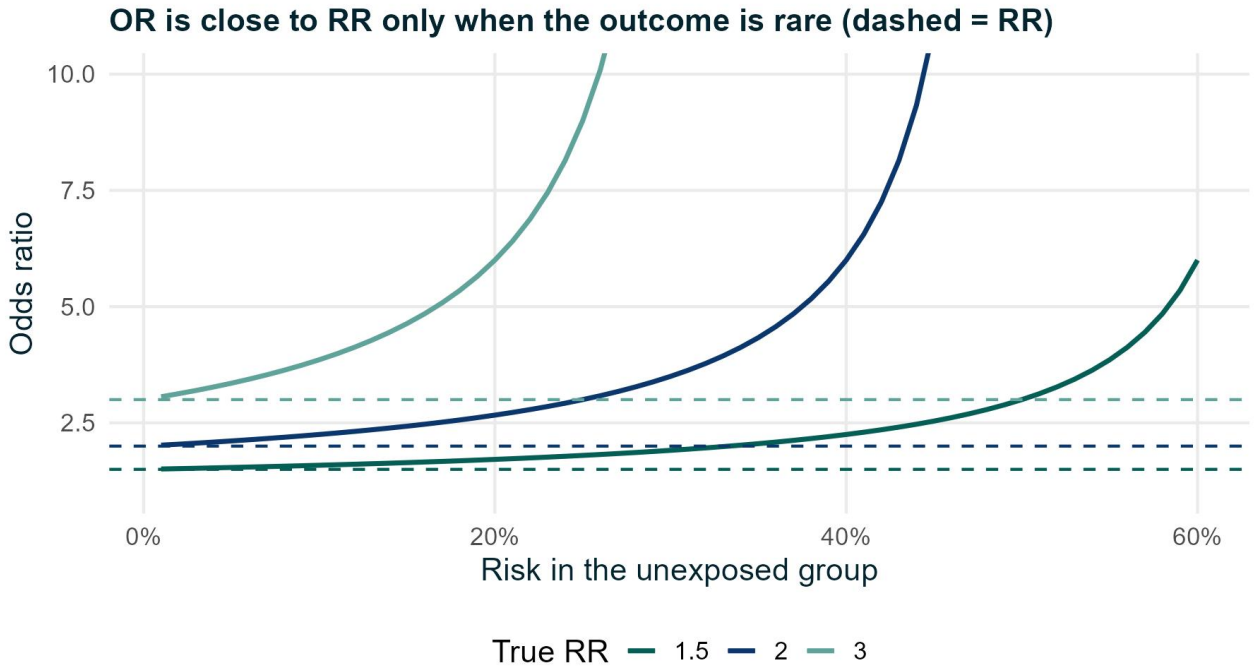
A 2 × 2 table: smoking and high blood pressure

	High BP: yes	High BP: no	Total	Proportion with high BP
Smoker	38	41	79	38 / 79 = 48.1%
Non-smoker	86	315	401	86 / 401 = 21.4%
Total	124	356	480	124 / 480 = 25.8%

- Row proportions answer: how common is high BP in each smoking group?
- Column proportions answer a different question: what share of people with high BP smoke?
- High BP here means SBP ≥ 140 mmHg at baseline (a cross-sectional measurement)

Proportions, risks and odds

- Proportion (prevalence): 38 / 79 = 0.48 vs 86 / 401 = 0.21
- Prevalence ratio (PR): 0.48 / 0.21 = 2.24 (95% CI 1.67 to 3.02)
- Odds: 38 / 41 = 0.93 vs 86 / 315 = 0.27
- Odds ratio (OR): 0.93 / 0.27 = 3.39 (95% CI 2.06 to 5.61)
- The OR is further from 1 than the PR because high BP is common



> Do not describe an OR of 3.4 as "3.4 times the risk". Here the prevalence is about 2.2 times higher.

The study design decides which measure you can interpret

Design	What you can estimate	Example
Cohort or randomised trial	Risks, risk difference, risk ratio (RR), OR	Adverse events by programme
Cross-sectional survey	Prevalence, prevalence ratio (PR), OR	High BP by smoking at baseline
Case-control study	Only the OR (cases and controls are sampled by outcome)	(not in our data)

- "Risk" means new cases over a period among people initially free of the outcome
- Logistic regression gives ORs whatever the design; interpret them with the design in mind

Chi-square and Fisher's exact test

Chi-square test

- Tests whether two categorical variables are associated
- Compares observed counts with the counts expected under no association
- Smoking and high BP: $\chi^2 = 23.1$, $p < 0.001$
- Approximation works when expected counts are about 5 or more

Fisher's exact test

- Exact test: no large-sample approximation
- Adverse events: 14 / 240 (5.8%) vs 3 / 240 (1.3%)
- RR 4.67 (95% CI 1.36 to 16.0), Fisher p = 0.011
- Few events give a very wide CI: the size of the increase is uncertain

> A significant chi-square for a larger table says the proportions differ somewhere. Look at the row proportions to see where.

A first look at confounding

Analysis	Odds ratio for smoking and high BP	95% CI
Crude (all participants)	3.39	2.06 to 5.61
Women only	2.75	1.17 to 6.43
Men only	3.28	1.73 to 6.23
Combined, adjusted for sex (Mantel-Haenszel)	3.09	1.85 to 5.15
Combined, adjusted for age group (Mantel-Haenszel)	2.57	1.51 to 4.37

- Smokers in these data are more often men and older; both raise blood pressure
- Part of the crude association reflects age and sex, not smoking
- Day 2: regression adjusts for several confounders at once

Paired categorical data: McNemar's test

- Same people classified twice: high BP at baseline and at 12 months
- Lifestyle group: 28 changed from high to not high; 6 changed from not high to high
- Only these discordant pairs carry information about change
- McNemar's test: $p < 0.001$
- A chi-square test on these data would wrongly treat the time points as independent groups

Summary: choosing a comparison

Outcome	2 independent groups	> 2 independent groups	Paired
Continuous, mean meaningful	Welch t-test	Welch ANOVA (+ adjusted pairwise)	Paired t-test
Skewed, positive	t-test on log scale, or Wilcoxon rank-sum	ANOVA on logs, or Kruskal-Wallis	Paired t on logs, or signed-rank
Ordinal	Wilcoxon rank-sum	Kruskal-Wallis	Wilcoxon signed-rank
Binary	Chi-square or Fisher; RR/OR with CI	Chi-square (or Fisher)	McNemar

> These are starting points, not automatic rules. The full decision diagrams are in the reference pack.

Practical session (2 hours)

Part	Script	Workbook	Time
A. Import and inspect, fix missing-value codes	01_import_and_inspect.R	1.1 to 1.3	25 min
B. Describe and plot	02_describe_and_plot.R	1.4 to 1.6	30 min
C1. Compare groups (continuous, paired)	03_compare_groups.R	1.7 to 1.9	35 min
C2. Categorical associations	04_categorical.R	1.10 to 1.11	20 min
Review and questions			10 min

- Open the .Rproj file first, then each script in turn
- Extensions E1.1 to E1.6 are for those who finish early



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